

Lojasiewicz Pandrosion Descent

Analytic residual energies, stable finite-slope steps, and convergence under gated corrections

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Abstract

Previous Pandrosion descent papers isolated conditional Lyapunov inequalities for stable finite-slope pairs and for gated higher-order corrections. This paper connects those inequalities to analytic convergence theory. Let $G : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ or $G : U \subset \mathbb{C}^n \rightarrow \mathbb{C}^n$ be analytic and set

$$\mathcal{E}(x) = \frac{1}{2} \|G(x)\|^2.$$

A Pandrosion sequence is called Lojasiewicz-admissible if it is bounded, satisfies sufficient decrease, and satisfies a relative-error estimate linking $\nabla \mathcal{E}$ to the accepted step. Under the classical Lojasiewicz gradient inequality near any cluster point, every admissible sequence has finite length and converges to a critical point of $\mathcal{E}$. If the cluster point is a regular zero of G , the sequence converges to that zero. We then show that stable finite-slope Pandrosion steps with line search satisfy the sufficient-decrease part, and that small gated finite-slope Halley corrections preserve admissibility. The resulting convergence theorem is intentionally conditional: it does not claim that every probe is good, but it gives a rigorous target for probe selection, atlas normalization, equal-value pole avoidance, and Halley gating.

Scope and status. This paper is a convergence bridge. Its theorems use standard analytic Lojasiewicz/Kurdyka-Lojasiewicz hypotheses stated explicitly, while the final global avoidance statement remains conjectural.

1 Analytic energy and Pandrosion steps

Let $G : U \rightarrow \mathbb{R}^n$ be real analytic. Complex analytic systems may be read as real analytic systems on $\mathbb{R}^{2n}$. Define the residual energy

$$\mathcal{E}(x) = \frac{1}{2} \|G(x)\|^2.$$

A finite-slope Pandrosion step at x_k selects a probe b_k , a finite-slope matrix $\mathcal{Q}_k = \mathcal{Q}_G(x_k, b_k)$, and solves

$$\mathcal{Q}_k \delta_k = -G(x_k).$$

The accepted iterate has the form

$$x_{k+1} = x_k + \lambda_k(\delta_k + e_k),$$

where $0 < \lambda_k \leq 1$ is a line-search parameter and e_k is an optional gated correction, for example a finite-slope Halley correction.

2 Admissible descent sequences

Definition 2.1 (Lojasiewicz-admissible sequence). *A bounded sequence $(x_k) \subset U$ is Lojasiewicz-admissible for $\mathcal{E}$ if there exist constants $\sigma > 0$, $L > 0$, and k_0 such that for all $k \geq k_0$,*

$$\mathcal{E}(x_{k+1}) + \sigma \|x_{k+1} - x_k\|^2 \leq \mathcal{E}(x_k), \quad (1)$$

$$\|\nabla \mathcal{E}(x_{k+1})\| \leq L \|x_{k+1} - x_k\|. \quad (2)$$

Remark 2.2. *The sufficient-decrease inequality is the Lyapunov part. The relative-error inequality says that accepted steps are not arbitrary energy-lowering jumps; the step length controls stationarity of the next point. This is the standard structure behind many analytic descent convergence theorems.*

Assumption 2.3 (Lojasiewicz gradient inequality). *At every cluster point x_* of the sequence, there are constants $c > 0$, $\theta \in [0, 1)$, and a neighborhood V of x_* such that*

$$|\mathcal{E}(x) - \mathcal{E}(x_*)|^\theta \leq c \|\nabla \mathcal{E}(x)\| \quad (x \in V).$$

3 Finite-length convergence

Theorem 3.1 (Lojasiewicz convergence of admissible Pandrosion sequences). *Let G be analytic and let (x_k) be a Lojasiewicz-admissible sequence for $\mathcal{E} = \frac{1}{2} \|G\|^2$. If the sequence has a cluster point x_* , then (x_k) has finite length in a neighborhood of x_* :*

$$\sum_{k \geq k_1} \|x_{k+1} - x_k\| < \infty$$

for some k_1 . Consequently the whole tail converges to x_* , and x_* is a critical point of $\mathcal{E}$.

Proof. The decrease inequality implies that $\mathcal{E}(x_k)$ is nonincreasing and converges to some value $\mathcal{E}_\infty$. Since x_* is a cluster point and $\mathcal{E}$ is continuous, $\mathcal{E}_\infty = \mathcal{E}(x_*)$. For all large k , the iterates lie in a Lojasiewicz neighborhood of x_* .

Set $\Delta_k = \|x_{k+1} - x_k\|$. From (1),

$$\sigma \Delta_k^2 \leq \mathcal{E}(x_k) - \mathcal{E}(x_{k+1}).$$

The Lojasiewicz inequality and the relative-error bound give

$$|\mathcal{E}(x_{k+1}) - \mathcal{E}_\infty|^\theta \leq c L \Delta_k.$$

Using the concavity of $t \mapsto t^{1-\theta}$ on positive t , one obtains the standard telescoping estimate

$$\Delta_k \leq C \left[(\mathcal{E}(x_k) - \mathcal{E}_\infty)^{1-\theta} - (\mathcal{E}(x_{k+1}) - \mathcal{E}_\infty)^{1-\theta} \right]$$

after increasing k_1 if necessary. Summing over k gives finite length. Hence the tail is Cauchy and converges to x_* . Finally, (2) and $\Delta_k \rightarrow 0$ imply $\nabla \mathcal{E}(x_*) = 0$. $\square$

Corollary 3.2 (Regular-zero convergence). *If the limit point x_* in Theorem 3.1 satisfies $G(x_*) = 0$ and $DG(x_*)$ is invertible, then the Pandrosion sequence converges to the regular zero x_* .*

Proof. The conclusion is immediate from Theorem 3.1. Invertibility identifies the zero as isolated and regular. $\square$

4 Stable finite-slope steps supply decrease

Definition 4.1 (Stable finite-slope direction). *At x , a finite-slope pair (x, b) is (K, μ) -stable if $\mathcal{Q} = \mathcal{Q}_G(x, b)$ is invertible,*

$$\|\mathcal{Q}^{-1}\| \leq K,$$

and the direction $\delta = -\mathcal{Q}^{-1}G(x)$ satisfies

$$\langle \nabla \mathcal{E}(x), \delta \rangle \leq -\mu \|G(x)\|^2.$$

Proposition 4.2 (Line-search sufficient decrease). *Assume $\nabla \mathcal{E}$ is Lipschitz with constant L_E on the relevant level set. If (x, b) is (K, μ) -stable and $\delta = -\mathcal{Q}^{-1}G(x)$, then every*

$$0 < \lambda \leq \frac{\mu \|G(x)\|^2}{L_E \|\delta\|^2}$$

satisfies

$$\mathcal{E}(x + \lambda\delta) \leq \mathcal{E}(x) - \frac{\mu}{2} \lambda \|G(x)\|^2.$$

If, in addition, the step sizes satisfy $\lambda \|G(x)\|^2 \geq c_0 \|\lambda\delta\|^2$, then the accepted steps satisfy sufficient decrease in the form (1).

Proof. The descent lemma gives

$$\mathcal{E}(x + \lambda\delta) \leq \mathcal{E}(x) + \lambda \langle \nabla \mathcal{E}(x), \delta \rangle + \frac{L_E}{2} \lambda^2 \|\delta\|^2.$$

The stability inequality bounds the linear term by $-\lambda\mu \|G(x)\|^2$, and the stated step-size bound absorbs half of it. The final statement follows by comparing the residual decrease with $\|\lambda\delta\|^2$. $\square$

5 Gated Halley corrections

The cubic finite-slope Halley layer should not be accepted unconditionally in a global convergence proof. It must be gated.

Definition 5.1 (Descent-compatible gate). *Let δ_k be the raw stable Pandrosion direction and h_k a higher-order correction. A gate is descent-compatible if the accepted correction $e_k = \omega_k h_k$ satisfies*

$$\|e_k\| \leq \eta \|\delta_k\|$$

with $\eta > 0$ small enough that the line search still accepts a sufficient-decrease step.

Proposition 5.2 (Gated corrections preserve admissibility). *Suppose the raw Pandrosion steps satisfy sufficient decrease and the relative-error condition on a bounded level set. If the gated corrections satisfy $\|e_k\| \leq \eta \|\delta_k\|$ with η sufficiently small and the line search is applied to the corrected direction $\delta_k + e_k$, then the corrected sequence is still Lojasiewicz-admissible, possibly with different constants.*

Proof. For small η , the corrected direction remains within a fixed cone around the raw descent direction:

$$\|\delta_k + e_k\| \asymp \|\delta_k\|.$$

The descent lemma therefore gives the same Armijo-type decrease after reducing the accepted step length by a uniform factor if needed. The relative-error bound is stable under this perturbation because the step norms are comparable and all derivatives are bounded on the level set. $\square$

6 Pandrosion convergence theorem

Theorem 6.1 (Conditional analytic convergence of gated Pandrosion). *Let G be analytic and let a Pandrosion engine generate a bounded sequence using stable finite-slope pairs, line search, and descent-compatible gated corrections. Assume the resulting accepted steps satisfy the relative-error estimate (2). Then every cluster point attracts the whole tail of the sequence, and the sequence converges to a critical point of $\mathcal{E} = \frac{1}{2} \|G\|^2$. If the cluster point is a regular zero, the sequence converges to that zero.*

Proof. Stable finite-slope pairs and line search give sufficient decrease by [Proposition 4.2](#). Gated corrections preserve admissibility by [Proposition 5.2](#). Analyticity gives the Lojasiewicz gradient inequality near cluster points. Apply [Theorem 3.1](#) and [Corollary 3.2](#). $\square$

7 Global conjecture

Conjecture 7.1 (Regular-attractor Pandrosion conjecture). *For analytic systems whose regular zeros are isolated, a dynamic Pandrosion engine combining stable probe selection, equal-value pole avoidance, m -flat atlas normalization, and descent-compatible finite-slope Halley gating avoids nonzero critical attractors for an open dense set of initial conditions.*

Remark 7.2. *This conjecture is stronger than the theorem. The theorem says that admissible trajectories converge. The conjecture says that the full engine can make the regular-zero alternative typical by avoiding singular probes, bad chart regions, and nonzero stationary traps.*

8 Conclusion

The Lojasiewicz viewpoint gives the convergence target for the Pandrosion programme. Geometry supplies stable charts and probes; line search supplies Lyapunov decrease; gates keep Halley corrections from destroying descent; analyticity converts these conditions into finite-length convergence. The theorem is conditional, but its conditions are exactly the design rules already isolated in the Pandrosion papers.

References

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